

Overview

This CAREER project will establish an integrated research and education program on adaptive artificial intelligence for real-time cybersecurity decision support in dynamic, adversarial communication environments. The motivating problem is that phishing and social-engineering attacks are shifting from static text messages toward interactive voice, video, and multimodal exchanges in which evidence is revealed sequentially and attacker behavior changes over time. Current detection systems are largely trained offline on static data and therefore provide limited scientific guidance for how AI systems should reason, adapt, and support human decisions under streaming, multimodal uncertainty.

The project has three objectives: (1) develop models for streaming multimodal inference when text, audio, and video signals arrive incrementally; (2) create self-updating learning methods that adapt to distributional drift and emerging attack strategies while controlling instability; and (3) study how human behavioral responses to adversarial communications can improve both AI decision support and cybersecurity training. The methods combine online representation learning, drift-aware model updating, multimodal threat detection, controlled human-subject experiments, and cyber-range training environments. Research findings will be incorporated into educational modules, and participant interactions with those modules will generate behavioral evidence used to refine model design and evaluation.

Intellectual Merit

The project advances knowledge by developing a principled framework for adaptive AI in environments where decisions must be made before complete information is available. Its first contribution is a formal treatment of partial temporal observability in multimodal cybersecurity settings, including representations that integrate sequential text, audio, and video cues under uncertainty. Its second contribution is a set of online adaptation mechanisms that identify evolving adversarial patterns while reducing catastrophic drift, overfitting to transient signals, and inappropriate updating from noisy feedback. Its third contribution is a human-centered account of adaptive decision support that links model behavior to measurable changes in user judgment, trust calibration, and susceptibility to deception.

Together, these contributions address a core limitation in current AI security research: the separation between model robustness, temporal adaptation, and human decision behavior. The expected results include new algorithms for streaming multimodal threat detection, empirical evidence about when adaptive updating improves performance relative to static or periodically retrained models, and validated methods for using behavioral signals in human-AI learning loops. Although phishing provides the testbed, the scientific advances generalize to broader classes of real-time decision systems operating under adversarial change, including fraud detection, misinformation response, and other high-stakes security workflows.

Broader Impacts

The project will benefit society by strengthening cybersecurity resilience against increasingly realistic AI-enabled phishing attacks in voice, video, and text communication channels. Improved detection and decision-support methods can reduce financial fraud, identity theft, organizational compromise, and misinformation harms. The project also advances desired societal outcomes through education, workforce development, broadening participation, and public access to research-based training resources.

Educational activities will produce modular cyber-range exercises, scenario libraries, and open instructional materials that can be adopted in undergraduate, graduate, and professional cybersecurity programs. The materials will be designed for use by Scholarship-for-Service institutions and partner educators preparing students for public-sector and national-security cybersecurity roles. Students involved in the project will receive interdisciplinary training in artificial intelligence, cybersecurity, human-centered computing, and experimental evaluation.

Broader impact outcomes will be assessed through measurable indicators, including learning gains, phishing-detection accuracy, changes in false-positive and false-negative decisions, participant confidence and trust calibration, module adoption, and usage across partner settings. Dissemination will include open-access educational materials, research publications, software artifacts when appropriate, and outreach to cybersecurity educators and workforce-development communities. By integrating research with scalable training infrastructure, the project will prepare a more capable cybersecurity workforce while producing scientific knowledge about adaptive AI systems that can serve the public interest.